ASSIGNMENT 2

SECI1143

SECTION 03

PROBABILITY & STATISTICAL DATA ANALYSIS

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Question 1

a) mean

midpoint	frequency	fX
155	12	1860
165	20	3300
175	5	875
185	3	555
	40	6590

$\bar{x} = \frac{6590}{40}$

$= 164.75$

b) Median

class interval	Frequency	Cumulative frequency
$150 \leq x < 160$	12	12
$160 \leq x < 170$	20	32
$170 \leq x < 180$	5	37
$180 \leq x < 190$	3	40

$N \div 2 = 40 \div 2$

$= 20$

$\therefore$ median class $= 160 - 170$

$L = 160$

$N = 40$

$cf_p = 12$

$W = 10$

$f_{med} = 20$

$\text{median} = L + \frac{\frac{N}{2} - cf_p}{f_{med}}(W)$

$= 160 + \frac{\frac{40}{2} - 12}{20}(10)$

$= 164$

c) Mode

$= l + h \times \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}$

$= 160 + 10 \times \frac{(20-12)}{(2(20) - 12 - 5)}$

$= 163.5$

d) Modal class

$= 160 \leq x < 170$

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Question 2

85, 90, 75, 88, 92, 80, 85, 82, 90, 85

a) mean $= \dfrac{85+90+75+88+92+80+85+82+90+85}{10}$

$= 85.2$

median $= \dfrac{85+85}{2}$

$= 85$

mode $= 85$

b) mean = gives average score of the students in the quiz

median = is the middle value of the quiz score, after arranging it in orders.

mode = is the value that occurs more frequently.

↳ mean represents the best summary of the score data because it shows the average value from the whole data set.

c) 55, 65, 65, 70, 85, 95, 95, 95, 100, 100

i) mean $= \dfrac{55+65+65+70+85+95+95+95+100+100}{10}$

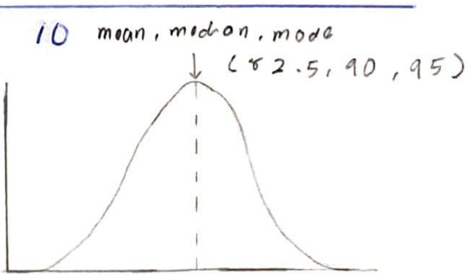
$= 82.5$

median $= \dfrac{85+95}{2}$

$= 90$

mode $= 95$

ii)



iii) Since the mean in this dataset is lower than the previous mean, it shows that the students performance is dropping.

Question 3

a) i) Range

$= 40 - 25$

$= 15$

ii) Variance

~~mean~~ $= \frac{384}{12}$

$= 32$

$$s^2 = \frac{\Sigma(x_i - \mu)^2}{N-1}$$

$$= \frac{230}{12-1}$$

$= 20.909$

iii) $s = \sqrt{20.909}$

$= 4.573$

b) Range of RM 15,000 shows the difference between the highest sale (RM45,000) and the lowest sale (RM 25,000)

Variance of 20.909 indicates that the sales of each month is dispersed from the average sales. Therefore, it shows there is variability in business monthly sales.

Standard deviation of 4.573 suggests that monthly sales figure deviate from the mean by an average of of approximately RM 4573.

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3 c) We can use variance and standard deviation of the sales to see variability of sales.

Greater variance and standard deviation shows high variability which can may cause challenges managing budgets, inventory and resource allocation.

We can also adjust marketing strategies for certain months to reduce variability in sales and identify opportunities for cost reduction.

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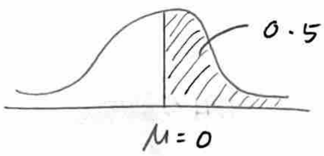
Question 4

$\mu = 50 \quad \sigma = 10 \Rightarrow X \sim N(50, 10)$

a) $p(x > 50)$

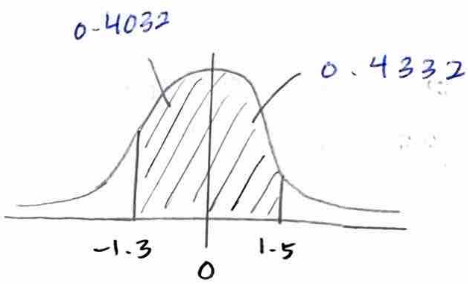


$z = \dfrac{x - \mu}{\sigma} = \dfrac{50 - 50}{10} = 0$

$0.5 \times 100 = 50\%$

from table : $z = 0.5$

b) $p(37 < x < 65)$



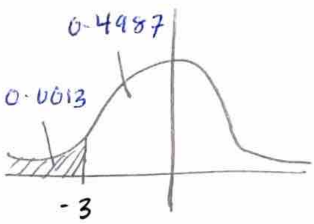
$z = \dfrac{37 - 50}{10}$ $\quad$ $z = \dfrac{65 - 50}{10}$

$= -1.3$ $\quad$ $= 1.5$

$= 0.4032 + 0.4332$

$= 0.8364$

c) $p(x < 20)$



$z = \dfrac{20 - 50}{10}$

$= -3$

$= 0.5 - 0.4987$

$= 0.0013$

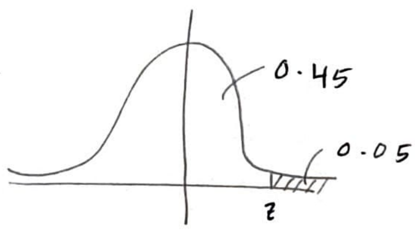
$x = 50 + (-3)(10)$

$= 20$

cost $= 20 \times 200$

$=$ RM4 000

d)



$0.45 \rightarrow z = 1.645$

$1.645 = \dfrac{X - 50}{10}$

$X = 66.45$

$\therefore$ So, minimum score need to be achieved is 66.45 units.

Question 5

a) Number of correct answers a student gets on exam.

b) $P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$

$P(x=0) = {}^{6}C_{0}\,(0.25)^{0}(0.75)^{6} = 0.178$

$P(x=1) = {}^{6}C_{1}\,(0.25)^{1}(0.75)^{5} = 0.356$

$P(x=2) = {}^{6}C_{2}\,(0.25)^{2}(0.75)^{4} = 0.297$

$P(x=3) = {}^{6}C_{3}\,(0.25)^{3}(0.75)^{3} = 0.132$

$P(x=4) = {}^{6}C_{4}\,(0.25)^{4}(0.75)^{2} = 0.033$

$P(x=5) = {}^{6}C_{5}\,(0.25)^{5}(0.75)^{1} = 0.004$

$P(x=6) = {}^{6}C_{6}\,(0.25)^{6}(0.75)^{0} = 0.0002$

x	0	1	2	3	4	5	6
$P(x)$	0.178	0.356	0.297	0.132	0.033	0.004	0.0002

c) $\mu = np$

$= 6(0.25)$

$= 1.5$

d) $P(x \geq 3) = 0.132 + 0.033 + 0.004 + 0.0002$

$= 0.169$

e) $P(X=4) = (1-p)^{x-1} p$

$= (1-0.75)^{4-1}(0.75)$

$= 0.012$

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Question 6

(a) $b^*(x;4,0.70) = \binom{x-1}{3}(0.70)^4(0.30)^{x-4}, \quad x = k, k+1, k+2, \ldots$

(b) $\sigma^2 = npq$

$\sigma = \sqrt{npq}$

$= \sqrt{4(0.7)(0.3)}$

$= 0.9165$

(c) $b^*(6,4,0.70) = \binom{6-1}{4-1}(0.70)^4(0.30)^{6-4}$

$= \binom{5}{3}(0.70)^4(0.30)^2$

$= 0.21609$

(d) $P(X=7) = \binom{12}{7}(0.70)^7(0.30)^5$

$= 0.1585$